Jinhang Wei(尉金行)

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Google Scholar

Education

- **Wenzhou University of Technology, China** *(Transfer to UMN)* ***Sept. 2021 – Nov. 2024***
  Bachelor's Program in Computer Science and Technology (Transferred)
  GPA: 3.77/4.0 (TOP5%)
  Awards: China National Scholarship **(awarded twice, top 0.2% nationwide** in 2022-2023 and 2023-2024**)**
- **University of Minnesota (Twin Cities), USA** ***Jan. 2025 – Dec. 2025***
  GPA: 3.94/4.0
  Bachelor's Program in Computer Science
- **University of Florida, USA** ***Jan. 2026***
  Ph.D. Program in Computer Science

Personal Profile

- **Research Achievements: Published 6 first-author Q1 SCI papers** in journals including J. Phys. Chem. Lett., BMC Biol., Brief Bioinform., and IEEE J. Biomed. Health, etc.
- **Research Interests:**
  Computational Biology, Artificial Intelligence in Life Sciences, Deep Learning
- **Previous Research Areas:** Artificial Intelligence-Driven Drug Discovery (AIDD), Graph Neural Networks, AI Disease Diagnosis, Applications of Large Language Models in Biology

Research Experience

1. Improving Neural Network Prediction Performance on Imbalanced Data through Data Resampling and Contrastive Learning Strategies

06/2022 – 06/2023

- **Drug-Disease Association Prediction Model:** Proposed a contrastive learning strategy based on dual perspectives of topology and semantics within GNN to address the over-smoothing phenomenon of heterogeneous nodes during message passing in drug-disease association datasets. This strategy improved the quality of node representations and achieved SOTA performance in downstream tasks. This work was published in ***Briefings in Bioinformatics.***
- **Graph Neural Network (GNN) Data Sampling Strategy:** Developed a strategy that effectively balanced the GNN's training process by repeatedly reconstructing ablation subgraphs of long-tail distribution datasets, significantly improving GNN's prediction accuracy on long-tail data. This work was published in ***Computers in Biology and Medicine.***
- **Drug-Food Association Prediction Model:** To address the challenge of accurately representing food data, we proposed a multi-view, multi-level masked contrastive learning strategy in this work, effectively improving the quality of data representation and enhancing prediction accuracy. This work was published in Computers in ***Computers in Biology and Medicine.***

2. Research on Drug Repositioning and Drug-Target Interaction Prediction Using Large Language Models and Deep Ensemble Learning

06/2023 – 10/2024

- **Drug-Target Interaction Prediction Model:** To enhance the reliability of computational predictions in wet lab experiments, we employed ensemble learning and a clustering-enhanced parameter fine-tuning method based on the distribution structure of drug (target) data, effectively addressing the false negative issue in datasets. Additionally, we designed a method for identifying key drug substructures based on Shapley values. Through model predictions, molecular docking, molecular dynamics simulations, and wet lab experiments, we successfully revealed previously undiscovered interactions between the target NQO1 and the drugs SIRT-IN-1 and LY2183240. This work was published in ***The Journal of Physical Chemistry Letters***, and was selected as a ***Cover Article.***
- **Drug Repositioning Framework Based on Large Language Models (LLMs):** To explore the potential of LLMs in drug discovery, we designed a framework based on LLMs that can accurately perform drug repositioning tasks according to target specificity. This framework comprehensively evaluates the

performance of mainstream LLMs in drug repositioning and conducts experiments comparing them with traditional deep learning models, quantifying and analyzing the unique advantages of LLMs. This work was published in ***BMC Biology.***

3. Diagnosis Model for Chronic Lymphocytic Leukemia and B-cell Acute Lymphoblastic Leukemia Based on Dynamic Flow Cytometry Data

11/2023 – 01/2024

- **Leukemia (Blood Cancer) Diagnosis Model:** We developed a leukemia diagnosis model based on a dynamic pooling method, which supports dynamic input of flow cytometry data. This model enables real-time monitoring of cancer cell levels in patients and significantly outperforms existing baseline methods, achieving a high level of diagnostic accuracy. Additionally, the diagnosis model has been successfully deployed on a web server, providing convenient online services for clinical application. This work was published in ***IEEE Journal of Biomedical And Health Informatics.***

4. GatorAffinity: Large-Scale Synthetic Structure-Affinity Database for AI-Driven Drug Discovery

01/2025 – NOW

• **Synthetic Structure-Based Protein-Ligand Affinity Database:** We developed GatorAffinity-DB, the largest synthetic database integrating experimental affinity labels (Kd/Ki) with deep learning predicted structures. Using the Boltz model, we generated 450K+ high-precision structure-affinity pairs from BindingDB and integrated 1M+ IC50 data from SAIR database, expanding existing datasets by tens of times. Pre-training our GatorAffinity model on synthetic data followed by fine-tuning achieved state-of-the-art performance on leak-proof benchmarks, surpassing all existing methods. We discovered a logarithmic data-scaling law, demonstrating that synthetic structures can effectively supplement scarce experimental data for drug discovery.

Published Papers

- **Wei, J.**, Zhang, Y., et al. **GatorAffinity: Boosting Protein-Ligand Binding Affinity Prediction with Large-Scale Synthetic Structural Data.** ***bioRxiv***, 2025.
- **Wei, J.,** Zhu, Y, Zhuo, L, et al. **Efficient Deep Model Ensemble Framework for Drug-Target Interaction Prediction.** ***The Journal of Physical Chemistry Letters***, 2024. (**Cover Article**, **Nature Index**, **JCR Q1**)
- **Wei, J.,** Zhuo, L., Fu, X, et al. **DrugReAlign: A Multisource Prompt Framework for Drug Repurposing Based on Large Language Models.** ***BMC Biology***, 2024. **(JCR Q1)**
- **Wei J.**, Zhuo L, Zhou Z, et al. **GCFMCL: predicting miRNA-drug sensitivity using graph collaborative filtering and multi-view contrastive learning**. ***Briefings in Bioinformatics***, 2023. (**JCR Q1**)
- **Wei, J.**, Wang, L., Zhou, Z, et al. **BloodPatrol: Revolutionizing Blood Cancer Diagnosis - Advanced Real-Time Detection Leveraging Deep Learning & Cloud Technologies.** ***Journal of Biomedical And Health Informatics***, 2024. **(JCR Q1)**
- **Wei J**, Zhuo L, Pan S, et al. **Headtailtransfer: an efficient sampling method to improve the performance of graph neural network method in predicting sparse ncrna–protein interactions**. ***Computers in Biology and Medicine***, 2023. (**JCR Q1**)
- **Wei J**, Li Z, Zhuo L, et al. **Enhancing drug–food interaction prediction with precision representations through multilevel self-supervised learning.** ***Computers in Biology and Medicine***, 2024. (**JCR Q1**)
- Wang R, Wang T, Zhuo L, **Wei J,** et al. **Diff-AMP: tailored designed antimicrobial peptide framework with all-in-one generation, identification, prediction and optimization**. ***Briefings in Bioinformatics***, 2024. (**JCR Q1**)
- Zhou Z, Liao Q, **Wei J**, et al. **Revisiting drug–protein interaction prediction: a novel global–local perspective**. ***Bioinformatics***, 2024. (**JCR Q1**)
- Zhou Z, **Wei J**, Liu M, Zhuo L, Fu X, Zou Q, et al. **AnomalGRN: deciphering single-cell gene regulation network with graph anomaly detection.** ***BMC Biology***, 2025, (**JCR Q1**)

Technical Skills

- **Deep Learning:** Python, PyTorch, PyTorch-Geometric, PyTorch-Lightning
- **Simulation Experiments:** Molecular docking (AutoDock Vina), Molecular dynamics simulation (Gromacs)
- **Data Analysis and Visualization:** Pandas, NumPy, Matplotlib, Seaborn